CSE2430 - Operating Systems
Homework Memory

Turn-in Instructions

No hand-in required. Questions are meant for self-directed exam preparation and can be discussed in the midterm Q&A session.

1 Page Tables

Consider a system in which the virtual address space is 64 bit, the page size is 4kB, and assume that an entry of a page table requires 4 bytes. The total amount of physical memory in the system is 1GB.

a) How much space would a simple single-level page table take?

b) How many levels would a multi-level page table need if we wanted to ensure that every individual page table fits into a single page of memory?

c) How many page table entries does an inverted page table need to have?

2 Contiguous Memory Allocation

Consider a process with a heap size of 10 MiB, initially all empty and unused. The process executes the following program:

```
#define KILOBYTE (1024)
#define MEGABYTE (1024 * KILOBYTE)

void *p0 = malloc(2 * MEGABYTE);
void *p1 = malloc(1 * MEGABYTE);
void *p2 = malloc(3 * MEGABYTE);
void *p3 = malloc(1 * MEGABYTE);
void *p4 = malloc(2 * MEGABYTE);
free(p1);
free(p3);
free(p0);
free(p2);
void *p5 = malloc(1 * MEGABYTE);
void *p6 = malloc(2 * MEGABYTE);
```

Write down the memory addresses (in MB, assuming that the heap starts at 0 MB) to which $p5$ and $p6$ point, when the best fit, worst fit and first fit memory allocation algorithms are used.

	Best fit	Worst fit	First fit
p5			
p6			

3 Virtual Memory Performance

On a system with a memory access time of 38 cycles on which the CPU runs at 3.2 GHz clock and the transfer of a page from hard disk takes 10ms, how large can the probability of a page fault be if we want the effective memory access time to be within 5% of the minimum?

4 Page Replacement

Assume the following trace of page requests:

a, b, c, d, a, b, e, a, b, c, d, e

How many page faults are generated using

a) a page cache of size 3 page frames and FIFO replacement policy,

b) a page cache of size 4 page frames and FIFO replacement policy,

c) a page cache of size 4 page frames and clock replacement policy.